Toward Path-Constrained Routing in Mixture-of-Experts Models: Extending OPT as an Operational Routing Framework

Abstract

Recent work has suggested that routing mechanisms in modular large language models may be interpreted as approximations of cognitive pathway selection processes. Within this perspective, the Ontology–Process–Trajectory (OPT) framework provides a structural account in which reasoning emerges from the propagation of signals between generative sources and stabilization sinks. Under such a formulation, expert routing in Mixture-of-Experts (MoE) architectures can be understood as an instance of pathway selection within a distributed computational system.

However, existing interpretations remain descriptive rather than operational. In current architectures, routing decisions are implemented through statistical gating functions learned via gradient-based optimization, without explicit structural constraints corresponding to pathway dynamics. As a result, while pathway-based descriptions offer a useful interpretive lens, they do not directly influence how routing is computed within the system.

This paper extends the OPT framework by introducing the concept of pathway-constrained routing. Instead of treating pathways as post-hoc descriptions of routing behaviour, pathway structures are incorporated directly into the routing process itself. In this formulation, routing decisions are defined as probability distributions over source–sink pathway units, and expert modules are aligned with stabilization mechanisms within this pathway space. Routing is therefore reinterpreted as the selection of pathways rather than the selection of expert indices.

The present work does not introduce a concrete training algorithm and does not provide empirical validation. Rather, it focuses on defining the structural components required to transform pathway-based interpretations into routing mechanisms, including the construction of pathway spaces, the definition of pathway generation functions, and the formulation of pathway-constrained routing distributions. A minimal implementation framework is outlined to demonstrate how such a formulation could be integrated into existing architectures.

Taken together, this work suggests a possible transition from unconstrained statistical gating toward structurally interpretable routing, and outlines a direction for developing

modular systems in which signal propagation pathways are explicitly represented as part of the model's internal computation.

1 Introduction

Routing mechanisms play a central role in modern modular neural architectures (Rosenbaum, Cases, Riemer, & Klinger, 2019; Ünal & Başçiftçi, 2022), particularly in Mixture-of-Experts models, where a routing function determines which expert modules are activated for a given input (Mu & Lin, 2025). Through this mechanism, routing effectively controls how information propagates through the system (Zhou et al., 2022). In most existing implementations, routing decisions are produced by learned gating networks that map input representations to distributions over expert indices. These mappings are typically obtained through unconstrained optimization, with the objective of improving task performance under a given loss function.

While such approaches have demonstrated strong empirical effectiveness, they provide limited insight into the internal structure of routing decisions (Chi et al., 2022; J. Liu et al., 2026). The learned mappings between inputs and experts are often opaque, and the resulting routing behaviour is difficult to interpret beyond post-hoc analysis. Although expert specialization may emerge during training, the principles governing expert selection remain implicit and are not directly represented within the model's architecture.

One possible direction for addressing this limitation is to reinterpret routing behaviour in structural terms. The Ontology–Process–Trajectory (OPT) framework provides such a perspective by modelling reasoning processes as pathways connecting generative sources and stabilization sinks (E. Liu, 2026a, 2026b). Within this framework, cognitive activity is not described solely in terms of representations or transformations, but in terms of signal propagation trajectories between distinct functional components. Each pathway corresponds to a structured mapping from a mode of signal generation to a mode of stabilization.

Under this interpretation, routing in modular neural systems can be viewed as selecting pathways within a structured signal space (E. Liu, 2026a). That is, instead of selecting an expert directly, the system may be understood as selecting a pathway that determines how an input signal is generated, transformed, and stabilized. However, in current architectures, this correspondence remains purely interpretive. Routing mechanisms do not explicitly encode pathway structures, nor do they impose constraints that reflect pathway dynamics. As a result, pathway-based descriptions do not influence the computational process that produces routing decisions.

This distinction highlights a gap between structural interpretation and computational implementation. While the OPT framework offers a way to describe routing behaviour in terms of structured pathways, existing models continue to implement routing as

unconstrained statistical selection. The central question that follows is whether pathway structures can be incorporated directly into routing mechanisms, so that routing decisions are generated within a structured space rather than interpreted after the fact.

In this work, we explore this possibility by extending the OPT framework toward an operational formulation of routing. We propose that routing can be defined as the selection of pathways within a predefined space of source–sink pairs, and that routing distributions can be constructed over these pathway units. Under this formulation, expert modules are no longer treated as independent computational entities, but are aligned with stabilization mechanisms corresponding to the sink components of pathways. Routing thus becomes a structured mapping from input signals to pathway activation distributions, rather than a direct mapping to expert indices.

The contribution of this work is therefore conceptual and structural. It provides a formalization of routing as pathway selection, introduces the notion of pathway-constrained routing distributions, and defines the components required to operationalize this formulation, including pathway spaces, pathway generation functions, and expert–sink alignment. In addition, a minimal implementation framework is outlined to demonstrate how these ideas could be integrated into existing architectures without requiring fundamental redesign.

At the same time, it is important to emphasize that the present work does not constitute an empirically validated model. No training algorithm is specified, and no experimental results are provided. The objective is not to claim immediate performance improvements, but to establish a structural framework that makes pathway dynamics explicit and potentially interpretable within modular neural systems. By shifting routing from an unconstrained statistical process to a structured pathway-based formulation, this work outlines a possible direction for future research into interpretable and internally structured computation.

2 From Routing Interpretation to Routing Mechanism

The OPT framework provides a structural interpretation of routing behaviour in modular neural architectures by modelling reasoning processes as pathways connecting generative sources and stabilization sinks (E. Liu, 2026b). Under this interpretation, routing distributions in Mixture-of-Experts models can be viewed as approximations of pathway activation distributions over a structured signal space (E. Liu, 2026a). That is, the selection of experts for a given input can be understood as reflecting an underlying selection of pathways through which signals are generated and subsequently stabilized.

However, this correspondence remains interpretive rather than operational. In existing architectures, routing decisions are produced by gating networks that learn statistical

mappings from input representations to expert weights (T. Liu, Blondel, Riquelme, & Puigcerver, 2024). These mappings are optimized to improve task performance, but they do not explicitly encode pathway structures, nor do they enforce constraints that reflect the dynamics of signal propagation described in the OPT framework. As a result, although routing behaviour may be retrospectively interpreted in terms of pathways, such interpretations do not influence how routing decisions are computed (Chen, Deng, Wu, Gu, & Li, 2022).

This distinction highlights a fundamental gap between structural description and computational implementation. While the OPT framework characterizes reasoning as structured signal propagation between functional components, current routing mechanisms remain grounded in unconstrained statistical optimization over expert indices (Nguyen, Tran, Nguyen, & Baraniuk, 2025). Consequently, the internal logic of routing remains difficult to interpret in structural terms, and pathway-based descriptions cannot be directly used to analyse or regulate routing behaviour.

The central question that follows is whether pathway structures can be incorporated directly into routing mechanisms. Rather than treating pathways as post-hoc explanations of routing outcomes, it becomes necessary to consider whether routing itself can be defined as the selection of pathways within a structured space. Such a shift would require replacing or augmenting conventional gating functions with a mechanism that generates routing distributions over pathway units.

Within the OPT framework, pathways are defined as mappings between generative sources and stabilization sinks (E. Liu, 2026a, 2026b). A routing mechanism consistent with this formulation would therefore operate over the space of source–sink pairs, rather than over expert indices alone. This redefinition transforms routing from a statistical selection problem into a structured mapping from input signals to pathway activation distributions. In this setting, routing decisions are no longer arbitrary assignments of inputs to experts, but instances of pathway selection within a predefined structural topology.

The following section develops a conceptual formulation of such pathway-constrained routing, specifying the structural components required to represent routing decisions in terms of pathway dynamics.

3 Pathway-Constrained Routing

3.1 Pathway Space

Within the OPT framework, reasoning processes are represented as pathways connecting generative sources and stabilization sinks. Let S denote the set of generative sources and T denote the set of stabilization sinks. The pathway space can then be defined as the Cartesian product

$$\mathcal{P} = S \times T$$

where each element $p = (s, t) \in \mathcal{P}$represents a possible pathway connecting a source unit $s$to a sink unit t. In this formulation, the routing problem is redefined over the pathway space $\mathcal{P}$, rather than over a set of expert indices.

Each pathway corresponds to a structured signal propagation trajectory within the system. By explicitly defining the pathway space, routing decisions can be expressed in terms of selecting among these trajectories, thereby introducing a structural topology into the routing process. This differs from conventional formulations in which routing operates over an unstructured set of experts without explicit representation of the relationships between them.

3.2 Pathway Generation Function

To operationalize pathway-based routing, it is necessary to define a mapping from input signals to pathway activation distributions. Let $x$denote an input representation. We define a pathway generation function

$$P(p \mid x), p \in \mathcal{P}$$

which represents the probability that pathway $p = (s, t)$is activated given input x.

This formulation replaces conventional gating functions, which produce probability distributions over expert indices, with distributions defined over pathway units. The pathway generation function therefore serves as the central mechanism for routing decisions within this framework.

In contrast to standard gating networks, which learn routing behaviour purely through statistical optimization, pathway generation functions may incorporate structural constraints derived from the OPT framework. For example, pathway activation may depend on interactions between source activation patterns and candidate stabilization processes, rather than on unconstrained mappings from input representations. In this sense, routing is mediated by the compatibility between generative and stabilizing components, rather than by direct assignment to expert modules.

The specific parameterization of the pathway generation function is not specified in the present work. Its role is defined at the level of structure: to transform input signals into routing distributions that reflect pathway selection rather than arbitrary expert weighting.

3.3 Pathway-Constrained Routing Distribution

Given a pathway distribution $P(p \mid x)$, routing decisions can be expressed as the selection of pathways rather than individual experts. In a modular architecture, expert modules can be associated with stabilization sinks, allowing routing to be mediated through pathway selection.

Under this interpretation, the routing process consists of generating a distribution over pathways and then mapping selected pathways to corresponding expert modules. Because each pathway includes a stabilization component, expert activation is determined by the sink element of the selected pathway. In this sense, routing becomes a structured process in which signals propagate through pathways before being resolved by associated experts.

This formulation introduces an intermediate layer between input representations and expert activation. Instead of directly mapping inputs to experts, the system first selects pathways within a structured space and then activates experts based on the stabilization components of those pathways. This two-stage interpretation allows routing behaviour to be analysed in terms of pathway distributions, providing a more interpretable representation of internal computation.

3.4 Expert–Sink Alignment

For pathway-constrained routing to be meaningful, expert modules must correspond to stabilization mechanisms within the pathway space. Let each expert $E_i$be associated with a sink unit $t \in T$. Under this alignment, experts function as computational realizations of stabilization processes.

This mapping introduces a structural relationship between expert specialization and pathway dynamics. Rather than treating experts as independent computational units, they can be interpreted as components within a stabilization landscape defined by the OPT framework. Routing decisions therefore determine not only which expert is activated, but also which stabilization mechanism is applied to the propagated signal.

Such an alignment suggests that expert specialization may emerge through repeated activation of particular pathways during training. Over time, experts associated with specific sinks may become increasingly specialized in stabilizing particular classes of signals, reflecting the structure imposed by pathway selection. Although this process is not modelled explicitly in the present work, it provides a conceptual basis for understanding how structural constraints may influence expert behaviour.

3.5 From Statistical Gating to Structured Routing

The pathway-constrained formulation outlined above suggests a conceptual transition from statistical gating to structured routing. In standard Mixture-of-Experts architectures, routing distributions are defined over expert indices and are learned through unconstrained optimization (Dimitri, Regina, & Alfonz, 2025). In contrast, pathway-constrained routing defines distributions over pathway units, introducing an explicit structural topology into the routing process.

This shift does not imply that pathway-constrained routing is immediately implementable or that it would necessarily improve performance. Rather, it establishes a structural framework in which routing decisions are grounded in interpretable pathway dynamics. By making pathway structure explicit, the formulation provides a basis for analysing routing behaviour at the level of signal propagation trajectories, rather than solely at the level of expert selection.

At the same time, the framework remains conceptual. No specific training procedure or optimization strategy is assumed, and the practical implications of enforcing pathway constraints are left for future investigation. The primary contribution of this section is therefore to define how routing can be reformulated as pathway selection within a structured space, thereby bridging the gap between interpretive models of cognition and operational mechanisms in neural architectures.

4 A Minimal Framework for Pathway-Based Routing

The pathway-constrained formulation described above is conceptual and does not assume a specific implementation. However, to demonstrate that the proposed structure can be integrated into existing architectures, it is useful to outline a minimal framework in which pathway-based routing can be realized without requiring fundamental changes to model design.

The objective of this framework is not to define a complete training procedure, but to illustrate how pathway representations, pathway activation, and expert selection can be operationalized within a standard neural architecture. In particular, the formulation aims to preserve compatibility with existing transformer or mixture-of-experts systems while introducing an explicit pathway layer into the routing process.

The proposed framework decomposes routing into four components: source representation, sink projection, pathway construction, and routing aggregation. These components correspond to different stages of pathway activation within the OPT formulation and together define a structured mapping from input representations to expert outputs.

4.1 Source Representation

Given an input representation x, the first step is to construct a representation over generative sources. This is achieved through a mapping

$$z_S = f_S(x), z_S \in R^{|S|}$$

where each dimension of $z_S$corresponds to the activation strength of a source unit. In the standard OPT formulation, the source set consists of four elements (internal abstraction, external abstraction, internal representation, and external representation), and therefore $z_S \in R^4$.

This representation can be implemented as a lightweight projection layer applied to the input embedding. The purpose of this step is to decompose the input signal into components corresponding to different modes of signal generation, rather than treating the input as a single undifferentiated representation.

4.2 Sink Projection

The second step is to project the source representation into a space of stabilization sinks. This is defined as

$$z_T = f_T(z_S), z_T \in R^{|T|}$$

where each dimension of $z_T$represents the activation tendency of a sink unit. In the standard formulation, the sink set consists of four elements (internal construction, external construction, internal recognition, and external recognition), so that $z_T \in R^4$.

This projection can be implemented as a learned transformation that maps source activations to stabilization tendencies. Conceptually, this step captures how different modes of signal generation give rise to candidate stabilization processes, introducing a dependency between source and sink components.

4.3 Pathway Construction

Given the source representation $z_S$and the sink representation z_T, a pathway activation matrix can be constructed as

$$P = z_S \otimes z_T$$

where $\otimes$denotes the outer product. The resulting matrix $P \in \mathbb{R}^{|S|\times|T|}$represents the activation strength of each pathway $\left(s_i \rightarrow t_j\right)$, with each element

$$P_{ij} = z_S^{(i)} \cdot z_T^{(j)}$$

corresponding to the activation level of the pathway from source s_i to sink t_j.

This construction explicitly represents routing as a distribution over pathway units. Unlike conventional gating mechanisms, which produce weights over expert indices, the pathway matrix encodes the joint activation of source–sink pairs, thereby introducing a structured representation of routing decisions.

The pathway matrix can be normalized to form a probability distribution over $\mathcal{P}$, or used directly as a set of routing weights, depending on the desired implementation.

4.4 Routing and Expert Aggregation

Routing decisions are then derived from the pathway activation matrix. Each pathway $(s_i \to t_j)$ is associated with an expert module E_{ij}, or, in a simplified formulation, with an expert corresponding to the sink t_j.

The output of the model can be defined as

$$y = \sum_{i,j} P_{ij} \cdot E_{ij}(x)$$

where $E_{ij}(x)$ denotes the output of the expert associated with pathway (i, j), and P_{ij} acts as a routing weight.

In practice, this aggregation can be implemented using either dense routing or sparse routing. For example, a top-k selection over pathway activations may be applied to reduce computational cost, analogous to sparse gating in standard MoE architectures. Alternatively, pathway weights may be used directly in a soft routing scheme.

Under this formulation, expert activation is no longer determined directly by input representations, but is mediated through pathway selection. The routing process therefore becomes a structured mapping in which input signals are first decomposed into source components, then projected into stabilization tendencies, and finally combined into pathway activations that determine expert usage.

4.5 Compatibility with Existing Architectures

The proposed framework can be integrated into existing transformer or MoE architectures with minimal modification (Mu & Lin, 2025). The source representation and sink projection can be implemented as additional lightweight layers, while the pathway construction can be computed efficiently using standard tensor operations.

Routing and aggregation can reuse existing infrastructure for expert selection and weighted combination.

Importantly, this formulation does not require replacing existing expert modules or changing their internal structure. Instead, it modifies the routing mechanism by introducing an intermediate pathway layer. This allows pathway-constrained routing to be explored as an extension of existing systems, rather than as a completely new architecture.

At the same time, the framework remains conceptual. The effects of pathway-based routing on performance, efficiency, and training stability are not evaluated here and require empirical investigation. The purpose of this section is therefore to demonstrate that pathway-constrained routing can be expressed in a form compatible with current neural architectures, providing a concrete starting point for future implementation and experimentation.

5 Implications for Interpretable and Self-Regulating Systems

While the OPT framework has been introduced here as a structural formulation for pathway-constrained routing, its properties also suggest broader implications for the design of interpretable and partially self-regulating computational systems. In particular, by redefining routing as pathway activation, the internal operation of a model can be expressed in terms of structured signal propagation rather than opaque expert selection.

In standard neural architectures, interpretability is typically approached through post-hoc analysis (Madsen, Reddy, & Chandar, 2022). Internal states are examined after training, and explanations are inferred from observed behaviour. In contrast, a pathway-based formulation introduces a representation in which routing decisions are expressed directly as selections within a structured pathway space. Under this view, each forward pass can be decomposed into a sequence of transformations involving signal generation, pathway selection, and stabilization. Such decomposition does not require external probing, but is instead embedded in the routing mechanism itself.

This structural representation enables the possibility of inspecting model behaviour at the level of pathway distributions. For example, different inputs may induce different pathway activation patterns, and these patterns can be compared across contexts to analyse how the system processes information. Similarly, unstable or inconsistent routing behaviour may be reflected as fluctuations in pathway distributions, providing a potential basis for diagnosing failure modes. While these possibilities are not explored empirically in the present work, they follow directly from making pathway structure explicit within the routing process (Sobrinho, 2003).

Beyond interpretability, the pathway formulation also suggests a possible shift in how system dynamics are evaluated and adjusted. In conventional training paradigms, learning is primarily guided by external objectives, such as supervised loss functions or reinforcement signals (Morales & Escalante, 2022). However, if system behaviour is expressed in terms of pathway activations, it becomes possible to define internal evaluation criteria based on properties of these activations. For instance, repeated oscillation between incompatible pathways may indicate instability in signal processing, while persistent concentration on a narrow subset of pathways may indicate reduced adaptability. Similarly, failure to converge toward a consistent stabilization pattern may reflect incomplete or conflicting processing dynamics.

Such observations suggest that pathway-based systems could, in principle, incorporate internal signals reflecting stability, consistency, or coherence of pathway activation. These signals could be used to guide adjustments in routing behaviour, complementing externally defined objectives. Importantly, this does not imply replacing standard optimization procedures, but rather augmenting them with structural criteria derived from the pathway representation. In this sense, learning may be viewed not only as optimizing outputs, but also as refining the internal organization of pathway dynamics.

A related consideration concerns the interpretability of systems that undergo continuous adaptation (Rouse & Morris, 1986). In many cases, as models become more specialized, their internal representations may become less accessible to human interpretation (Hassija et al., 2024). Within the pathway-based formulation, this issue may be addressed by introducing a mapping between internal pathway activations and a structured representation that can be inspected externally. Because pathways are defined over a fixed set of source–sink units, their activation patterns can be represented in a consistent and interpretable form, even if the underlying model parameters evolve.

Such a mapping can be understood as an interface between internal computation and external analysis. The internal system may operate using high-dimensional representations optimized for performance, while pathway activations provide a lower-dimensional structural description of how signals are processed. This separation allows system complexity to increase without necessarily sacrificing the ability to monitor and analyse its behaviour at the level of pathway dynamics.

Taken together, these considerations suggest that pathway-constrained routing may support a class of systems in which internal computation is both structured and partially observable. Rather than treating interpretability as an auxiliary objective, it becomes a property of the routing mechanism itself. At the same time, the possibility of defining internal evaluation signals based on pathway dynamics raises the prospect of systems that can regulate aspects of their own behaviour in response to structural conditions.

It is important to emphasize that these implications remain speculative. The present work does not provide empirical evidence that pathway-based routing leads to improved interpretability, stability, or adaptability. Instead, it identifies a set of structural properties that follow from representing routing decisions as pathway

activations, and suggests how these properties might be explored in future research on modular and interpretable neural systems.

6 Discussion

The framework presented in this work proposes a shift in how routing mechanisms in modular neural architectures are conceptualized. Rather than treating routing as a direct mapping from inputs to expert indices (Mu & Lin, 2025), it is reformulated as the selection of pathways within a structured space defined by generative sources and stabilization sinks. This perspective introduces an explicit topology into the routing process, enabling routing decisions to be expressed in terms of signal propagation trajectories.

One implication of this reformulation is that routing behaviour can be analysed at a level that is not directly accessible in standard architectures (Hassija et al., 2024). Because pathway activations are explicitly represented, it becomes possible to study how different inputs induce different patterns of pathway selection, and how these patterns relate to downstream outputs. This may provide a more structured basis for interpreting model behaviour, compared to approaches that rely on analysing expert activations in isolation.

At the same time, the proposed framework raises several open questions. The definition of pathway units, as well as the choice of source and sink sets, is currently fixed for conceptual clarity. It remains unclear how these sets should be defined or adapted in large-scale systems, or whether more flexible or learned pathway spaces would be required in practice. Similarly, the relationship between pathway-based routing and existing optimization frameworks, such as standard gradient-based training or reinforcement learning, is not specified and would require further investigation.

Another consideration concerns the potential trade-offs introduced by pathway constraints. While structured routing may improve interpretability, it may also restrict the flexibility of routing decisions, potentially affecting performance (Rosenbaum et al., 2019). The extent to which such constraints can be introduced without degrading model capability remains an empirical question. In addition, the computational cost associated with representing and operating over pathway spaces, particularly in large-scale systems, would need to be evaluated.

Despite these limitations, the pathway-constrained formulation provides a clear conceptual framework for exploring structured routing mechanisms. By making pathway structure explicit, it offers a basis for connecting interpretive models of cognition with operational mechanisms in neural architectures. Whether this connection can be realized in practice remains an open question, but the formulation establishes a direction for investigating how structural constraints may be integrated into routing processes.

7 Conclusion

This paper extends the OPT framework from a descriptive account of cognitive pathways to a structural formulation for routing in modular neural architectures. By redefining routing as the selection of pathways between generative sources and stabilization sinks, it introduces a pathway-constrained perspective in which routing decisions are expressed over a structured space of source–sink pairs.

Within this formulation, routing distributions are defined over pathway units, pathway generation functions map inputs to pathway activations, and expert modules are aligned with stabilization mechanisms. A minimal implementation framework is outlined to demonstrate how these components may be integrated into existing architectures without requiring fundamental redesign.

The proposed framework remains conceptual and is not accompanied by a concrete training algorithm or empirical validation. Its primary contribution lies in defining a structural approach to routing that makes pathway dynamics explicit and potentially interpretable. By shifting the focus from expert selection to pathway instantiation, the framework suggests a direction for developing systems in which internal computation can be represented, analysed, and potentially regulated at the level of signal propagation.

Future work is required to evaluate whether pathway-constrained routing can be implemented efficiently, whether it supports improved interpretability in practice, and how it interacts with existing training paradigms. Nevertheless, the formulation presented here provides a starting point for exploring structured routing mechanisms in modular neural systems.

8 References

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